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CE205
Databases and Information Retrieval

Business Intelligence: Data Warehouses and Data Mining

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Lecturer

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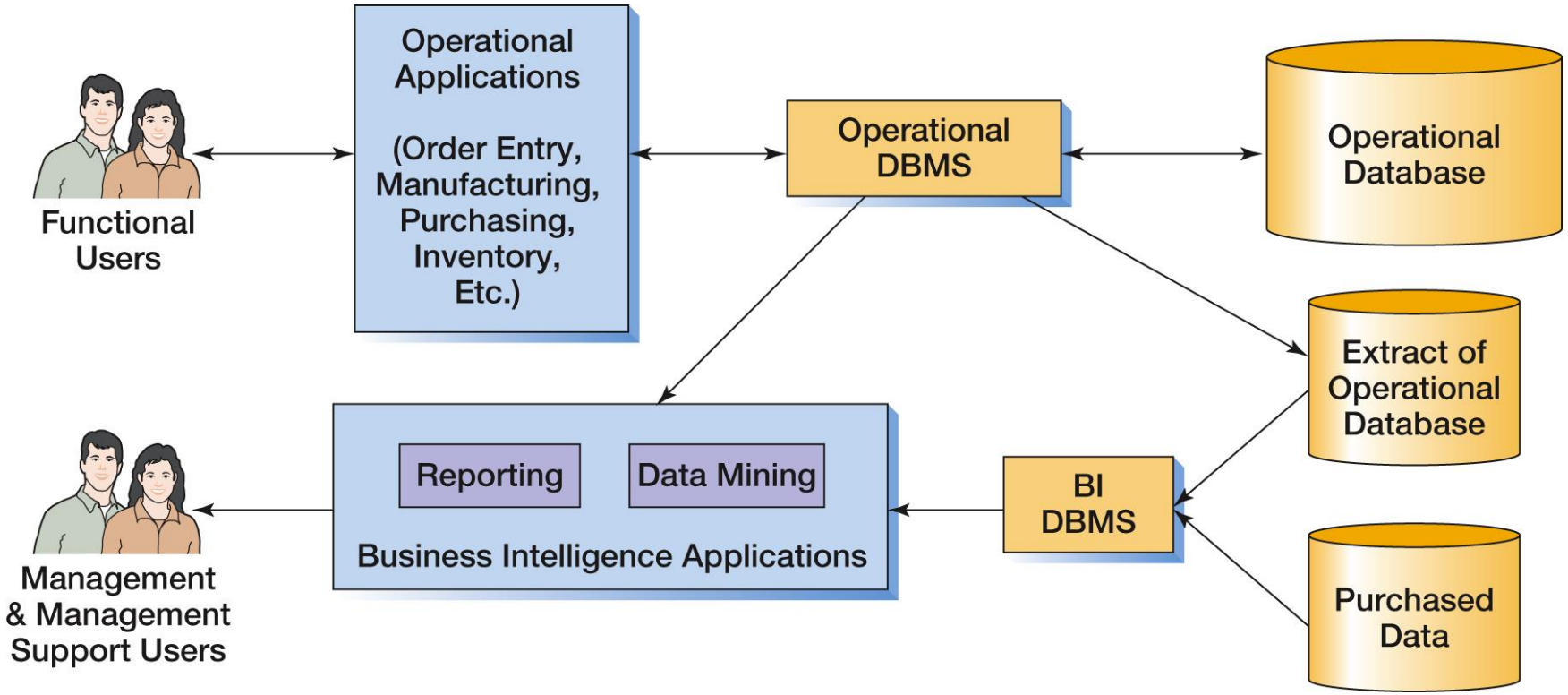
Most of the slides based on

- Database processing: fundamentals, design, and implementation (David M. Kroenke; David J. Auer) . 14th Edition (Global Edition). Pearson.

Learning Objectives

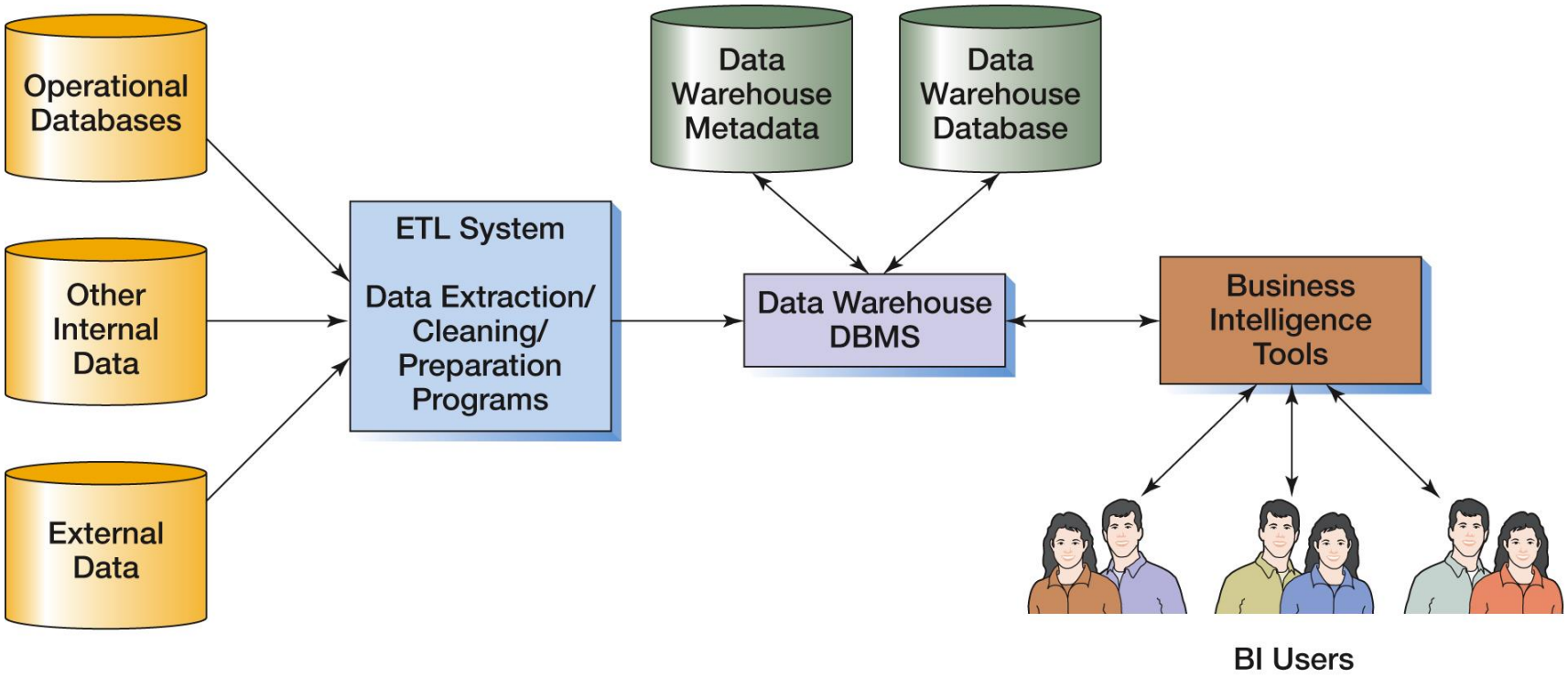
- What is Online Transaction Processing System (OLTP) and Business Intelligent System (BI)
- How BI can be related to operational systems
- What are data warehouses and marts and why we need them
- How data warehouse databases can be designed
- Dimensional Databases and Star Schema
- Reporting systems

Relationship Between Operational and BI Systems

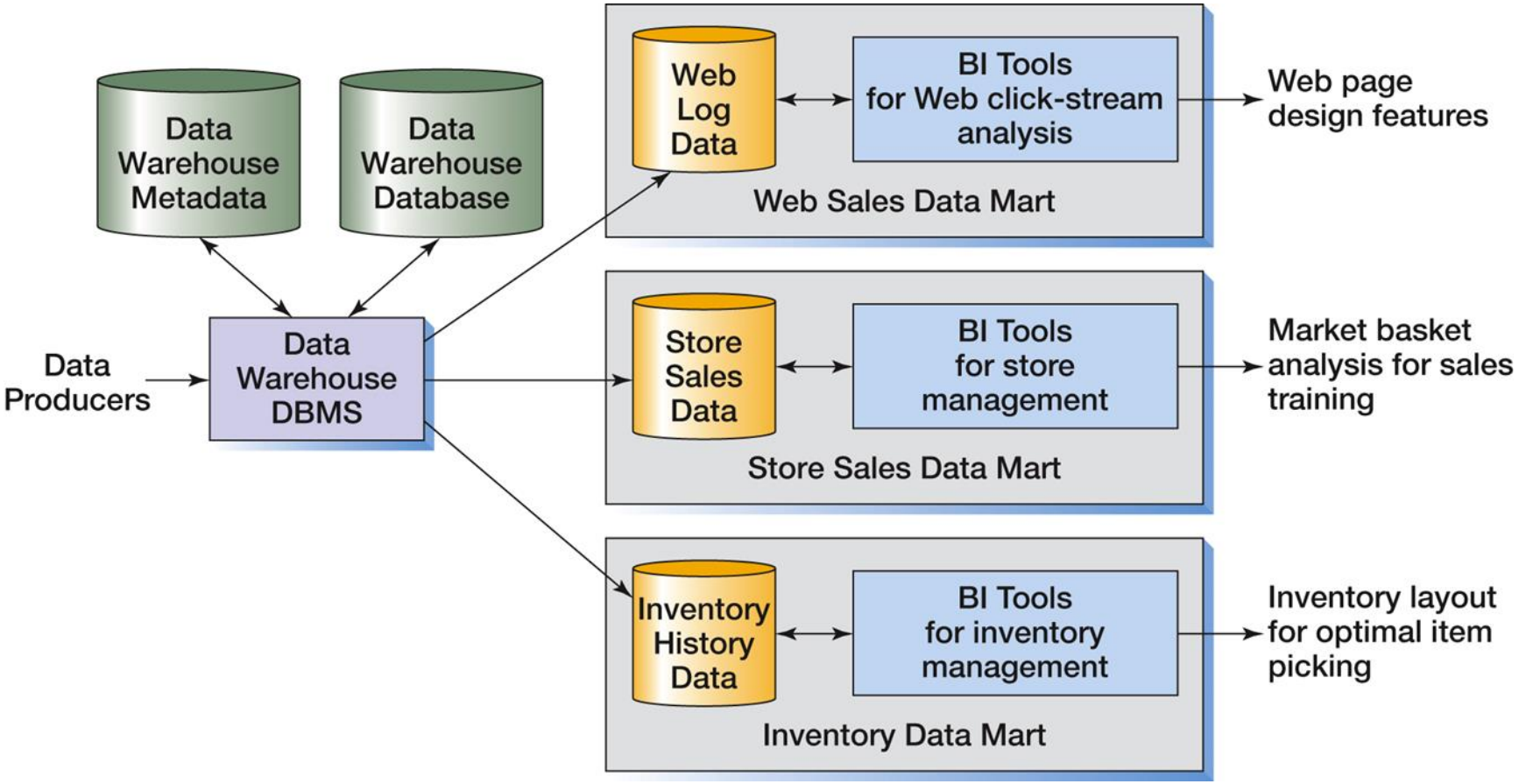


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Components of a Data Warehouse

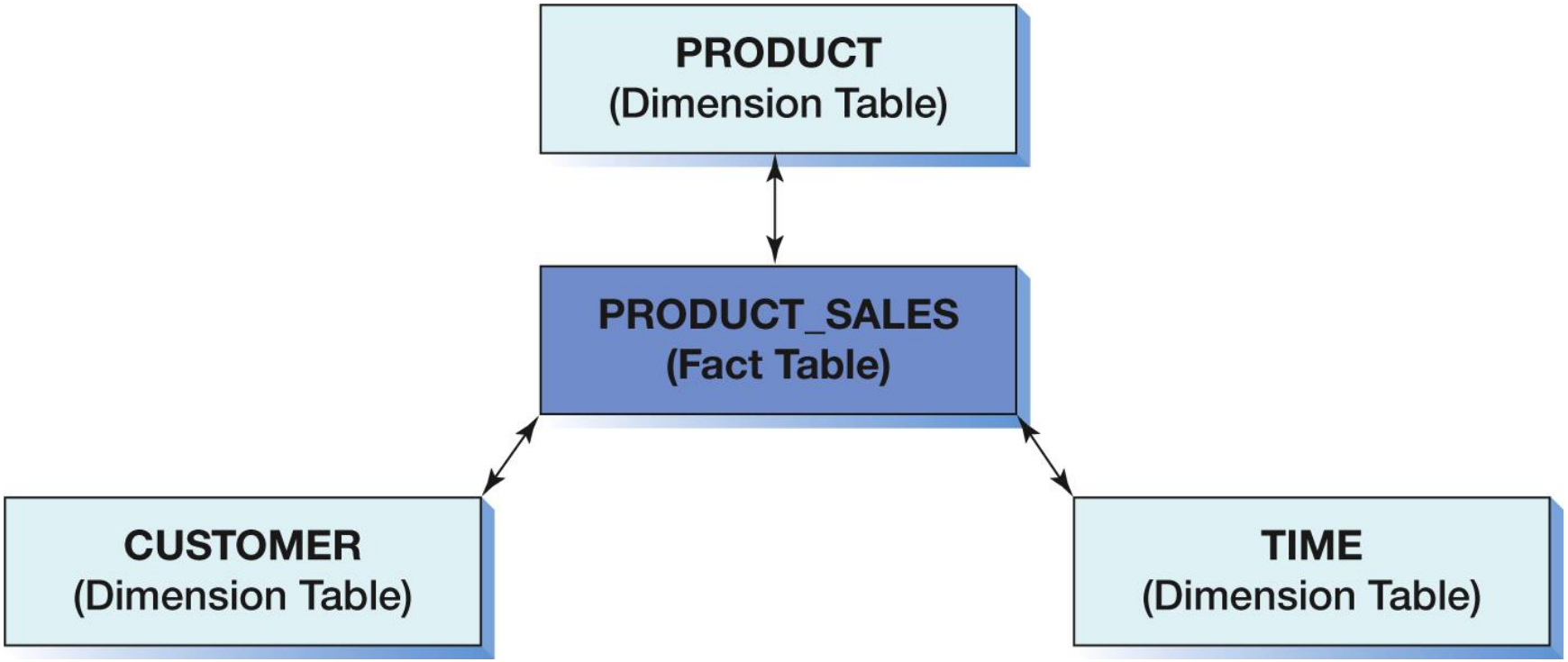


Data Warehouse and Data Marts



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Star Schema

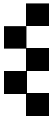

PRODUCT
(Dimension Table)
PRODUCT_SALES
(Fact Table)
CUSTOMER
(Dimension Table)
TIME
(Dimension Table)

RFM Score Report

Each customer is ranked for **R** (recent), **F** (frequent), and **M** (money) characteristics—1 is highest (best) and 5 is lowest (worst) score

	RFM Score		
Customer	R	F	M
Able, Ralph	1	1	2
Baker, Susan	2	2	3
George, Sally	3	3	3
Tyler, Jenny	5	1	1
Jacobs, Chantel	5	5	5

Real-time Data and its Volume



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Real-time Detection of Broadband Issues

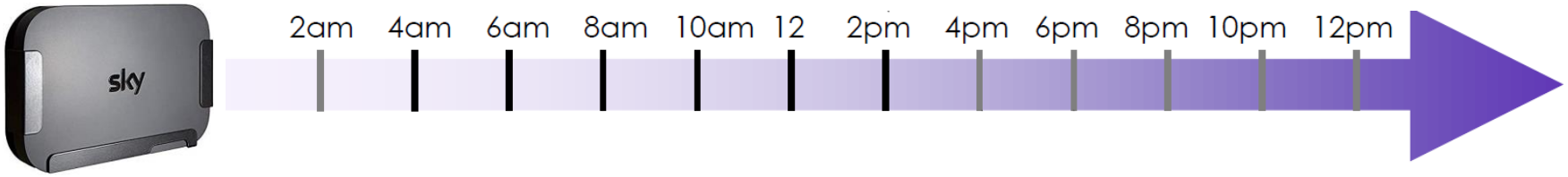
Poor Broadband performance can cause

- Bad customer experience
- Increase customer churn
- Bad repute for a brad in the market
- Increase in management cost

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Volume of Data

- Telecom companies have million of users
- If data is collected after every 2 hours to predict the broadband issues
- What will be the size of data?

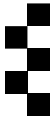


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Problems with Data

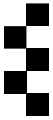
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Operational Systems and Business Intelligence Systems



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Online Transaction Processing System

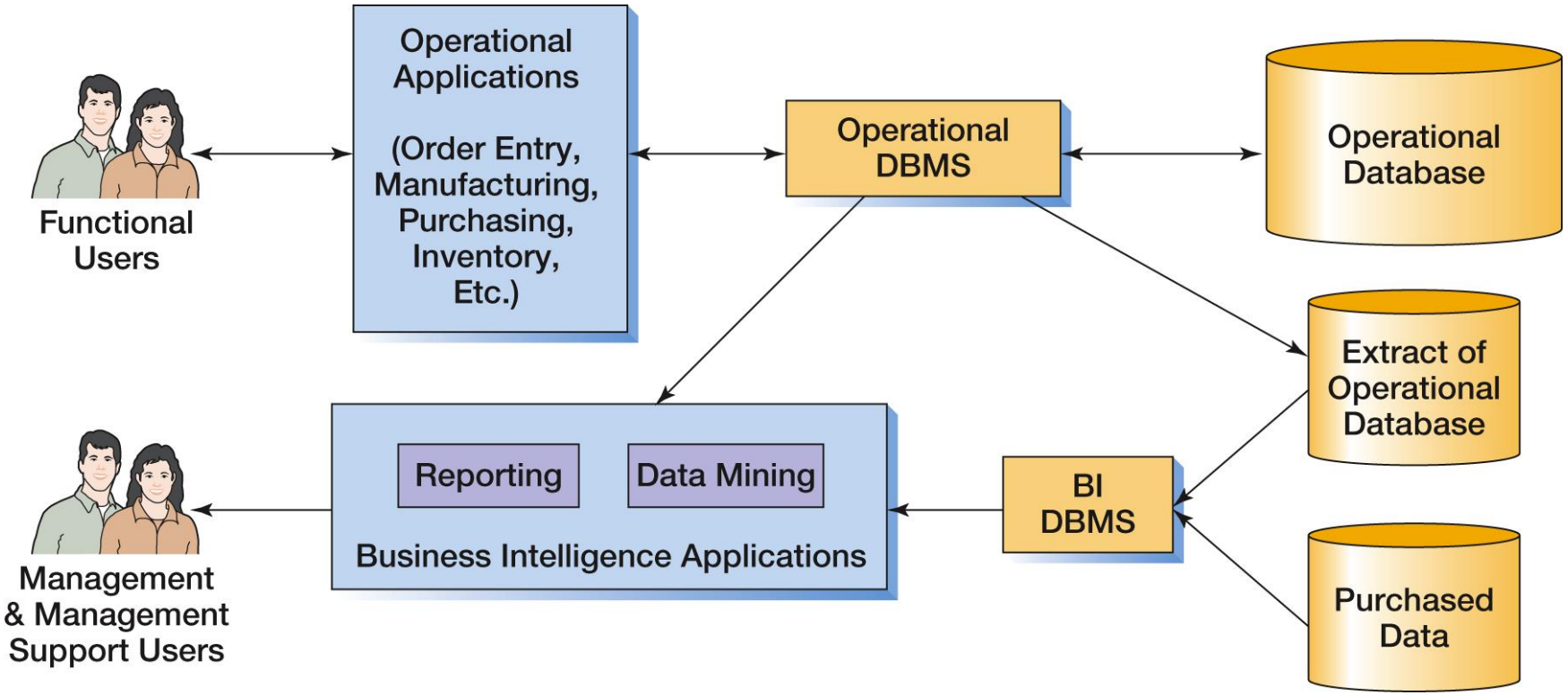
- An online transaction processing (OLTP) system is used to **record all sales transactions** of the company (whether in a store, on the Web, or from mail order or phone order sales).

- OLTP systems are the **backbone of businesses** as they operate today.

Business Intelligence Systems

- Business intelligence (BI) systems are information systems that **assist managers and other professionals** in the analysis of current and past activities and in the prediction of future events.
- Unlike transaction processing systems, they **do not support operational activities**, such as the recording and processing of orders.
- Instead, BI systems are used to support management assessment, analysis, planning, control, and, ultimately, **decision making**.

Relationship Between Operational and BI Systems (1)



Relationship Between Operational and BI Systems (2)

- Operational systems such as
  - sales
  - Purchasing
  - inventory control systems

support primary business activities.

- They use a DBMS to both read data from and store data in the operational database.
- They are also known as transactional systems or online transaction processing (OLTP) systems because they record the ongoing stream of business transactions.

Relationship Between Operational and BI Systems (3) Ways to Access Data by a BI System

BI systems obtain data from **three** possible sources.

1. **First**, BI systems read and process data existing in the operational database
   - they use the operational DBMS to obtain such data, but they do not insert, modify, or delete operational data.
2. **Second**, BI systems process data that are extracted from operational databases.
   - they manage the extracted database using a BI DBMS, which may be the same as or different from the operational DBMS.
3. **Finally**, BI systems read data purchased from data vendor.

Data Access from Operational Database

- Some BI systems **read and process operational data** directly from the operational database.
- Although this is possible for simple reporting systems and small databases, such direct reading of operational data is **not feasible** for more **complex applications** or **larger databases.**
- Those **larger applications** usually **process** a **separate database** constructed from an extract of the operational database.

Difficulties in Direct Accessing of Data from Operational Database

Operational data are difficult to read for several reasons.

- First, querying data for BI applications can place a substantial burden on the operational DBMS and unacceptably slow the performance of operational applications.
- Additionally, operational data have problems that limit their use for BI applications.

Requirement for BI Systems

- Further, the creation and maintenance of BI systems require
  - programs
  - facilities
  - and expertise
- Because of these problems, many organizations have chosen to develop **data warehouses and data marts** to support BI applications.

BI Support in DBMS

Oracle UK / Business Analytics / Business Intelligence

Business Intelligence (BI)—Discover New Insights

Are you a current Oracle Business Intelligence customer fostering a data-driven culture with powerful, visually stunning analytics? The Oracle Analytics platform has empowered people across the organization to make faster, more informed, and mobile-enabled business decisions. Now the platform has driven the current release of industry-leading cloud analytics, Oracle Analytics.



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